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(54) **VACUUM PUMP AND ROTATING BODY FOR VACUUM PUMP**

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CPC ***F04D 19/042*** (2013.01)

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CPC F04D 19/04; F04D 19/042; F04D 29/266
See application file for complete search history.

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(57) **ABSTRACT**

A vacuum pump which can reduce a stress generated in a connected part between a rotating body and a rotating shaft is provided. A rotating body is provided having rotor blades provided on an outer periphery of a rotor-blade forming portion, and fastened to the rotor shaft by a bolt, and rotatable with the rotor shaft. At least either one of a fitting hole portion fitted with the rotating shaft and a through hole portion, through which a bolt penetrates, in the rotating body is a stress-reduction target portion, and a groove portion that reduces stress generated in the stress-reduction target portion during rotation of the rotating body is provided.

9 Claims, 11 Drawing Sheets

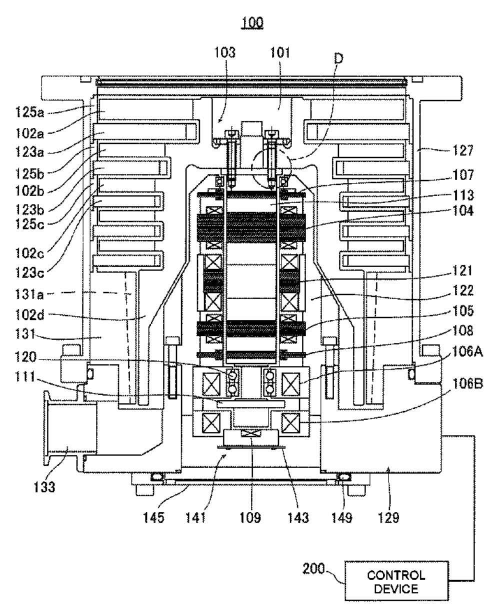

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Fig.1

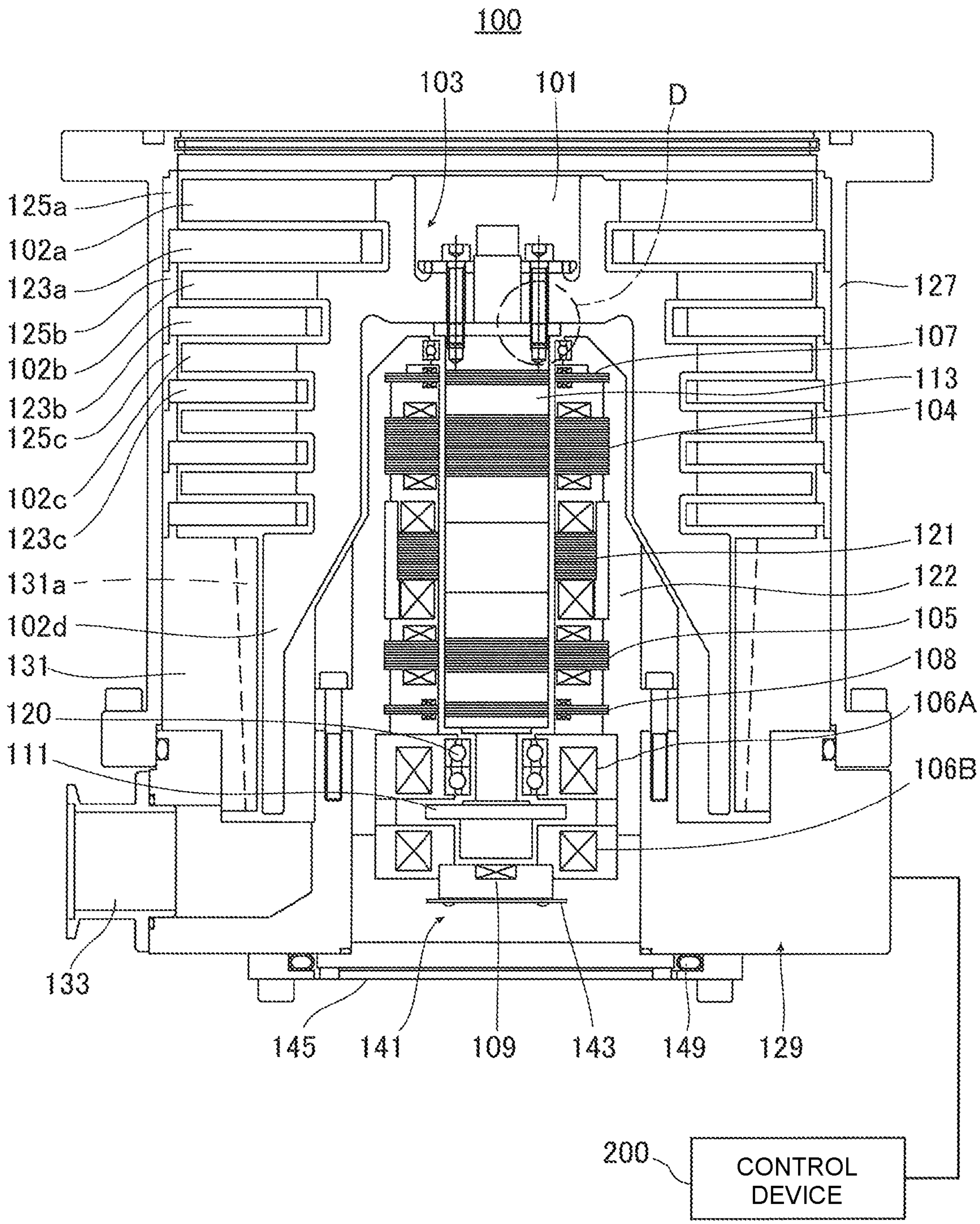
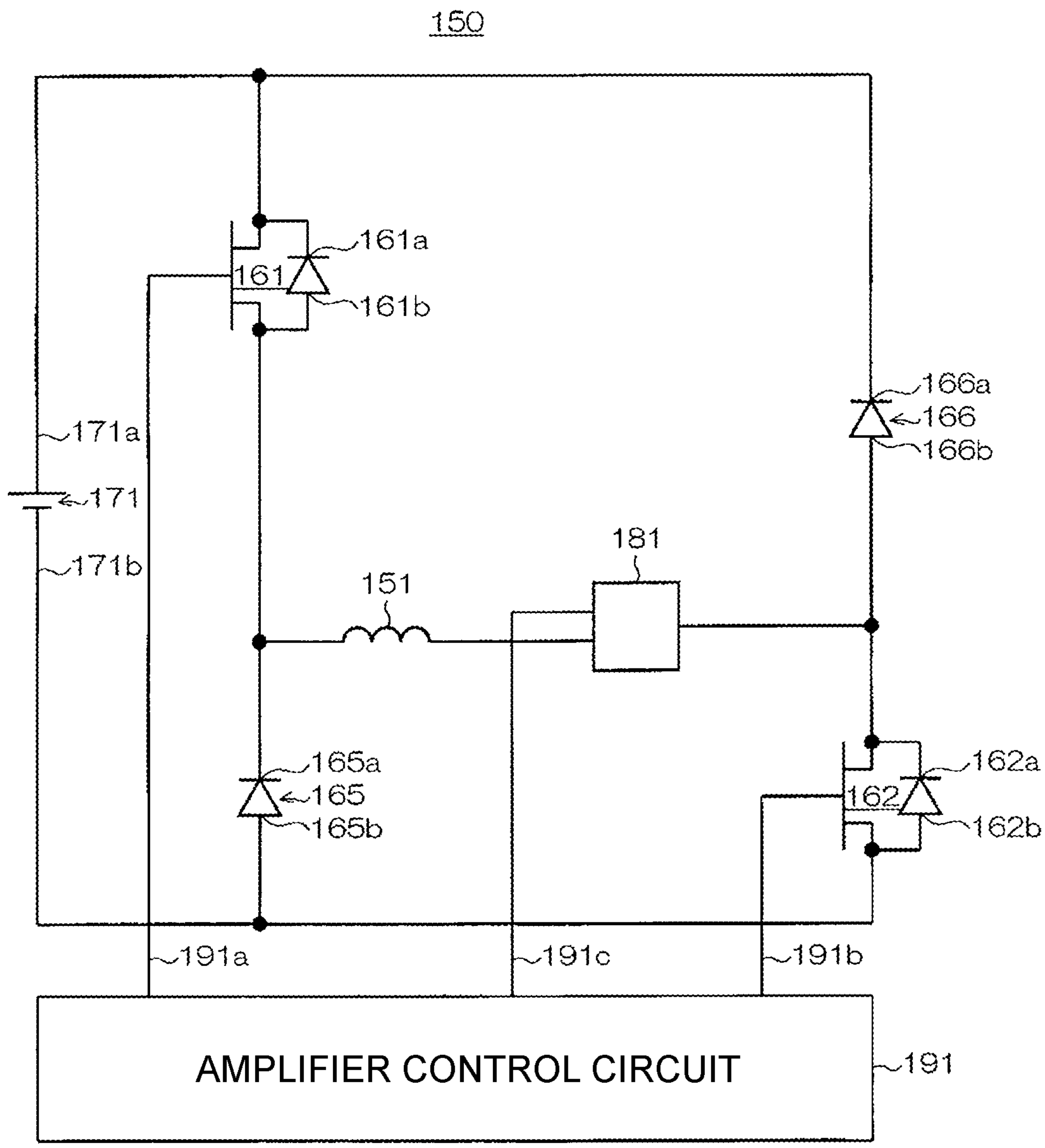

100
103
101
D
125a
102a
123a
125b
102b
123b
125c
102c
123c
131a
102d
131
120
111
127
107
113
104
121
122
105
108
106A
106B
133
145
141
109
143
149
129
200
CONTROL
DEVICE

Fig.2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161a
161
161b
166a
166
166b
171a
171
171b
181
151
165a
165
165b
162a
162
162b
191a
191c
191b
AMPLIFIER CONTROL CIRCUIT
191

Fig.3

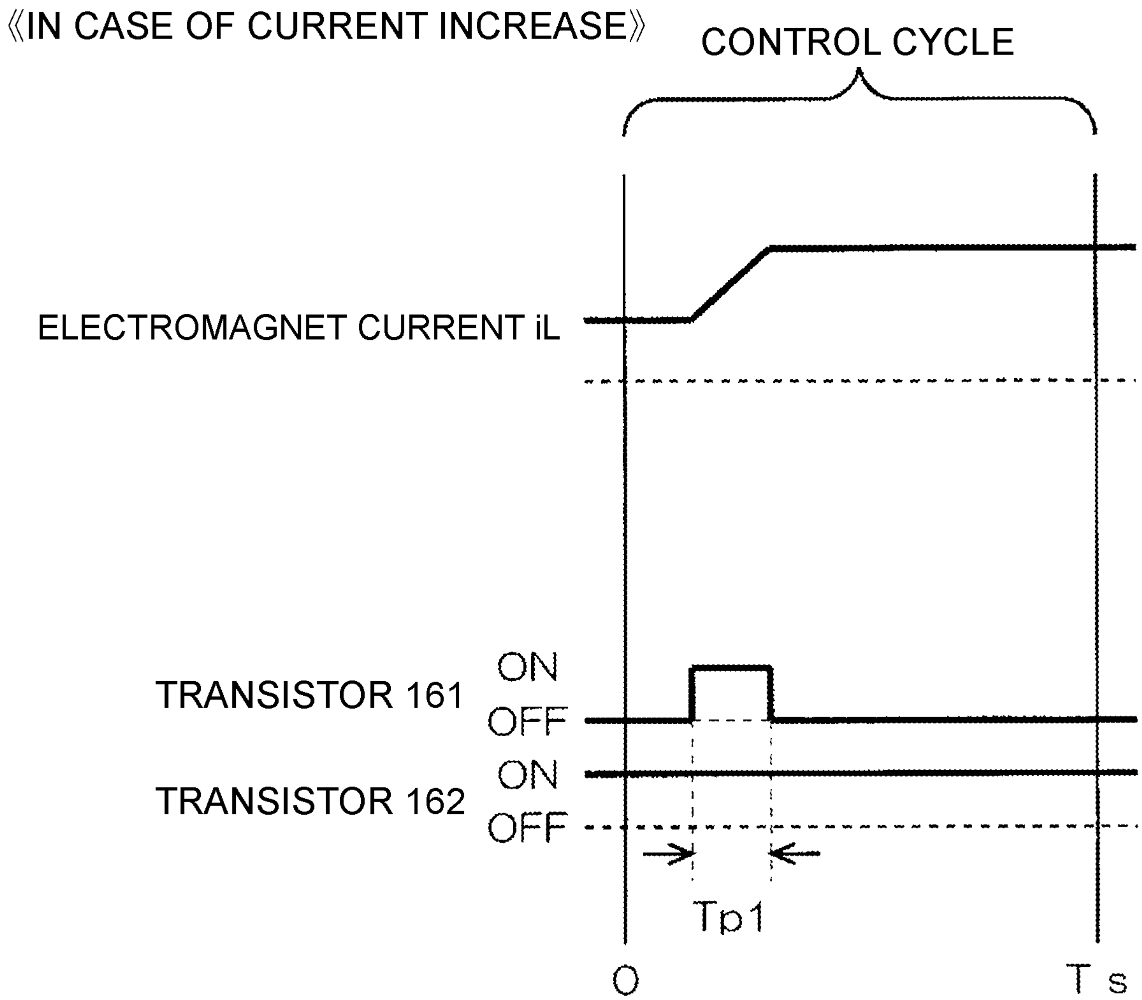

《IN CASE OF CURRENT INCREASE》
CONTROL CYCLE
ELECTROMAGNET CURRENT iL
TRANSISTOR 161
ON
OFF
TRANSISTOR 162
ON
OFF
Tp1
0
Ts

Fig.4

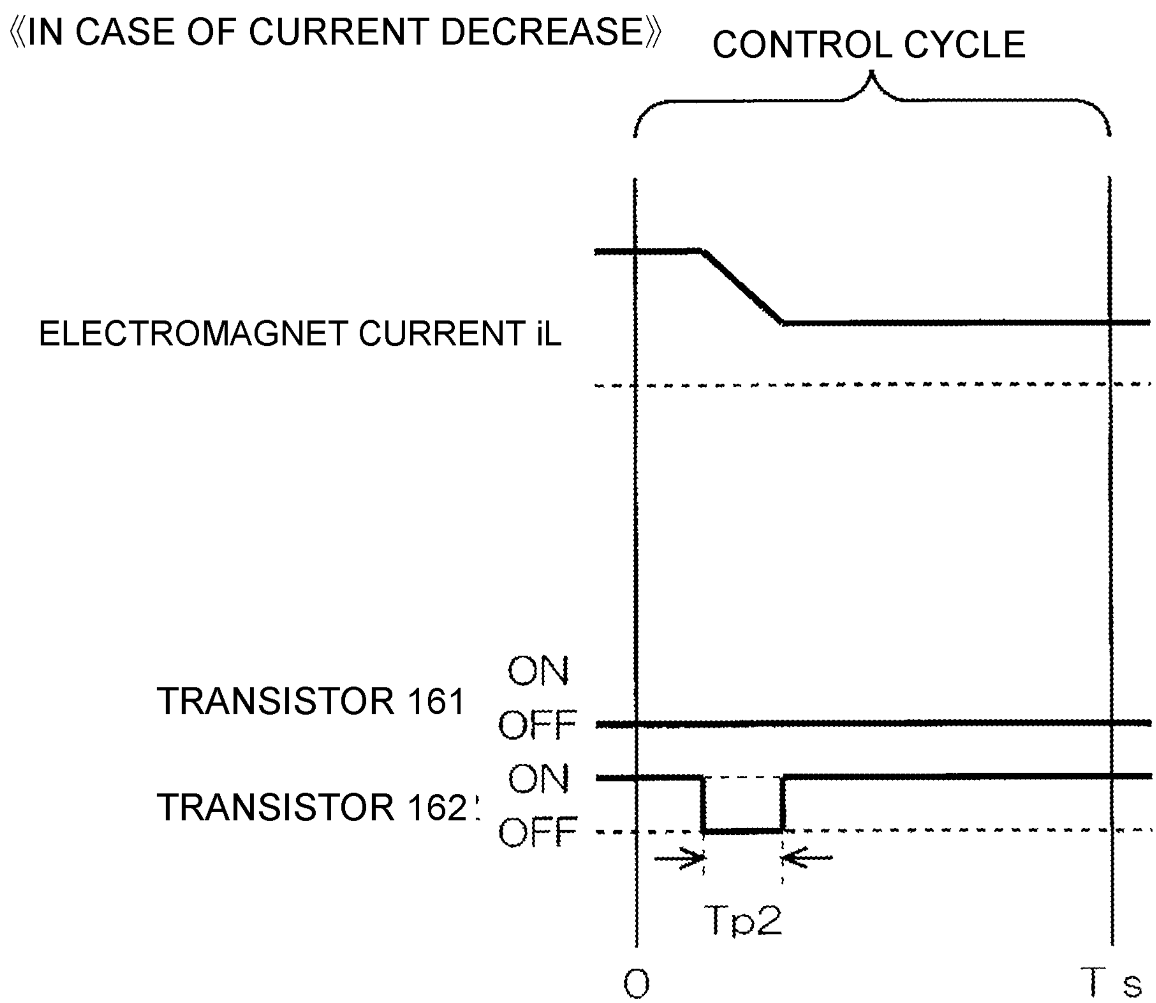
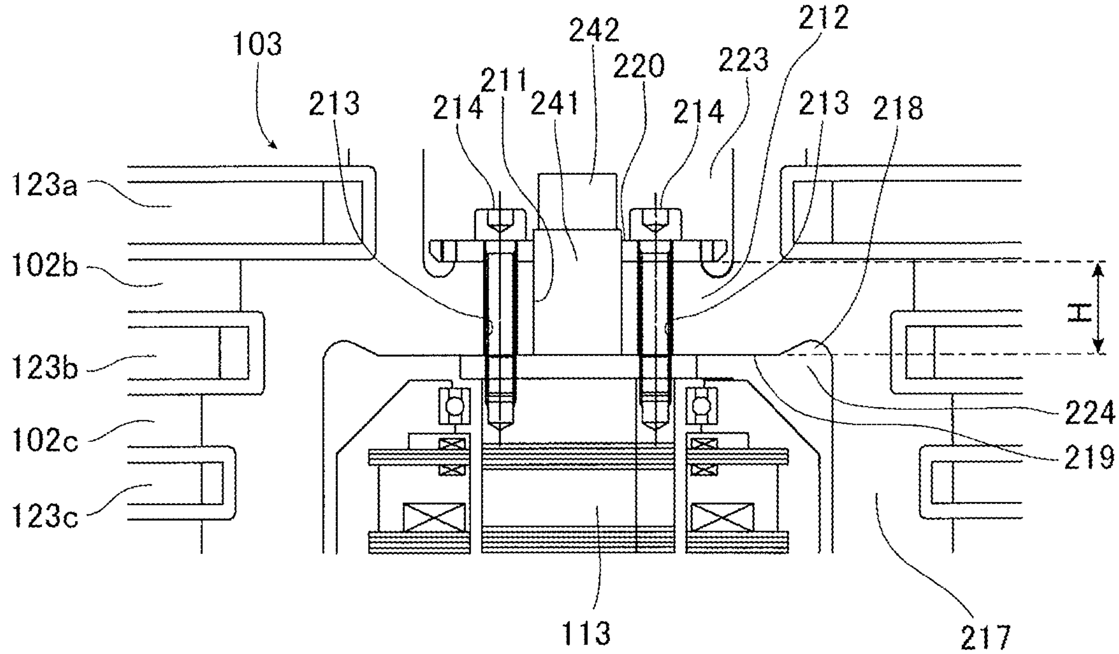

《IN CASE OF CURRENT DECREASE》
CONTROL CYCLE
ELECTROMAGNET CURRENT iL
TRANSISTOR 161
ON
OFF
TRANSISTOR 162
ON
OFF
Tp2
0
Ts

Fig.5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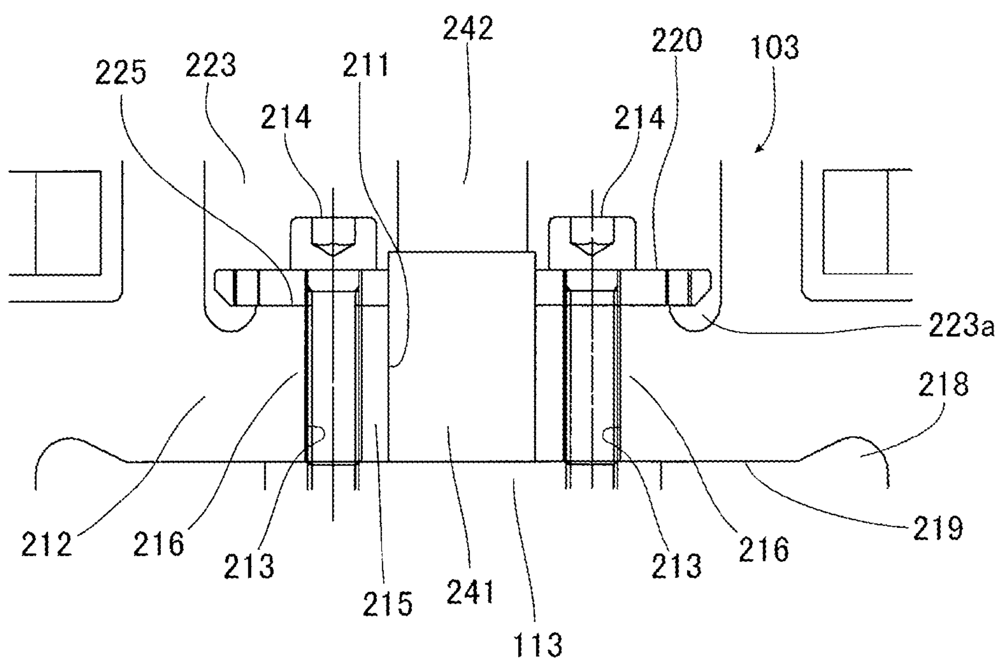
242
212
211
220
223
213
214
241
214
213
218
123a
102b
H
123b
224
219
102c
123c
113
217

Fig.6


242
220
103
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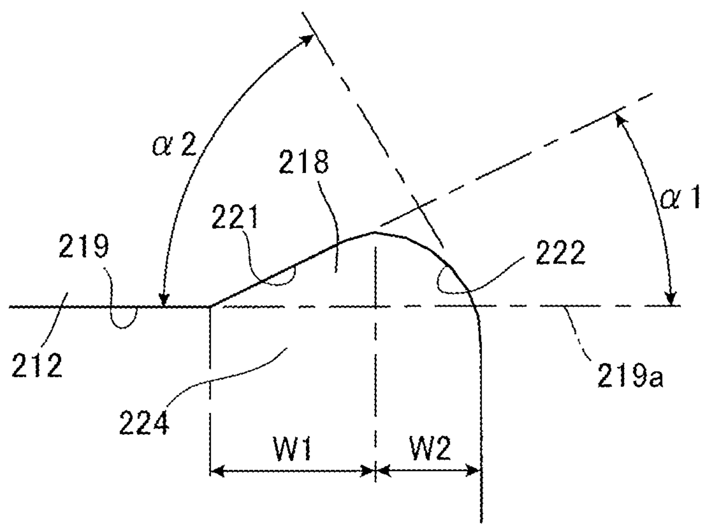
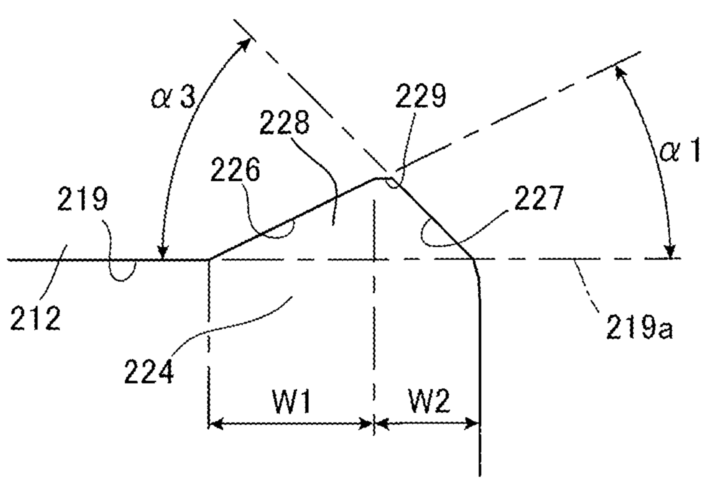
223
211
214
214
223a
218
212
216
213
215
241
113
213
216
219

Fig.7

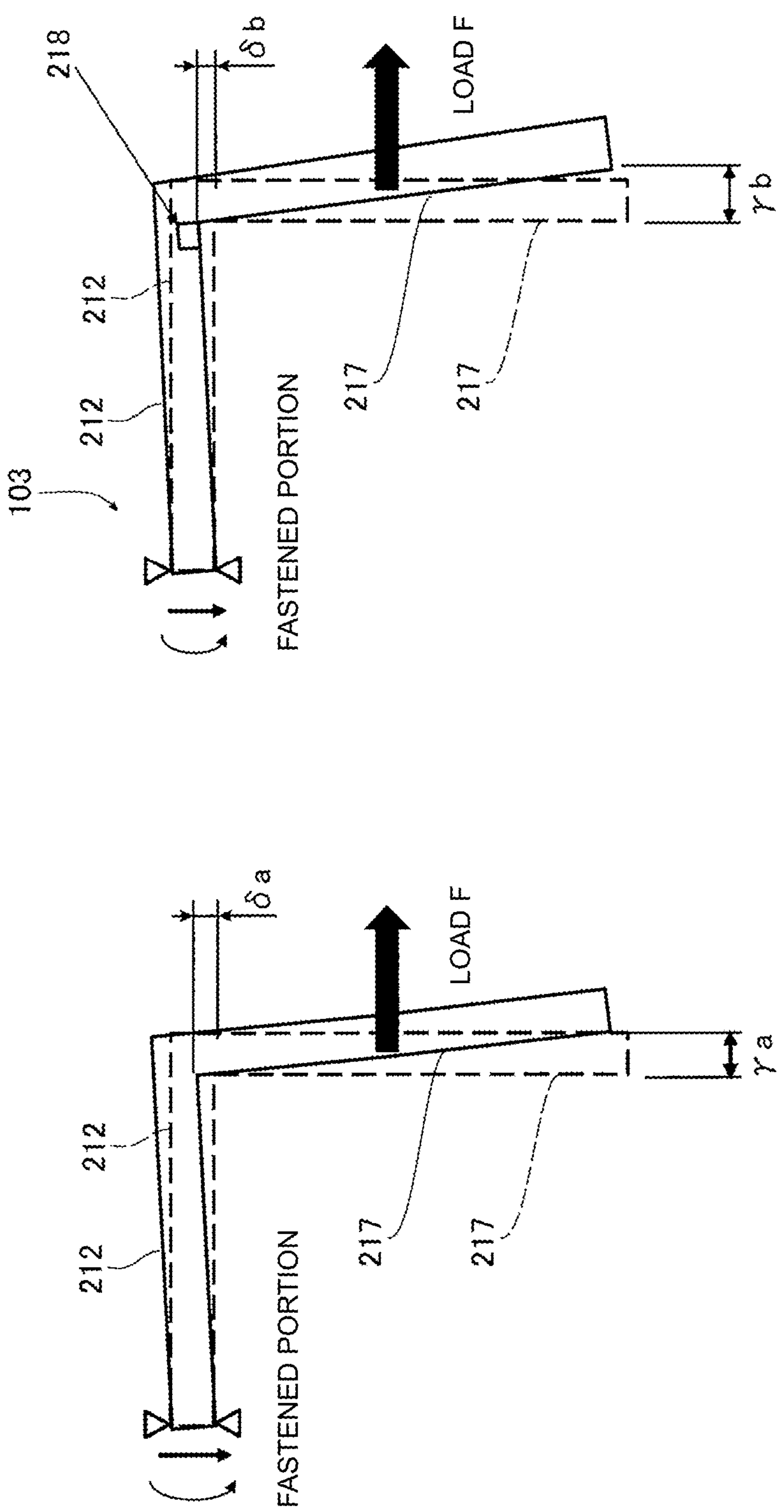


α2
218
221
219
222
α1
212
219a
224
W1
W2

Fig.8



α3
229
228
226
219
227
α1
212
219a
224
W1
W2



(b) STRUCTURE MODEL ACCORDING TO EMBODIMENT

(a) CONVENTIONAL STRUCTURE MODEL

FIG. 9

Fig.10

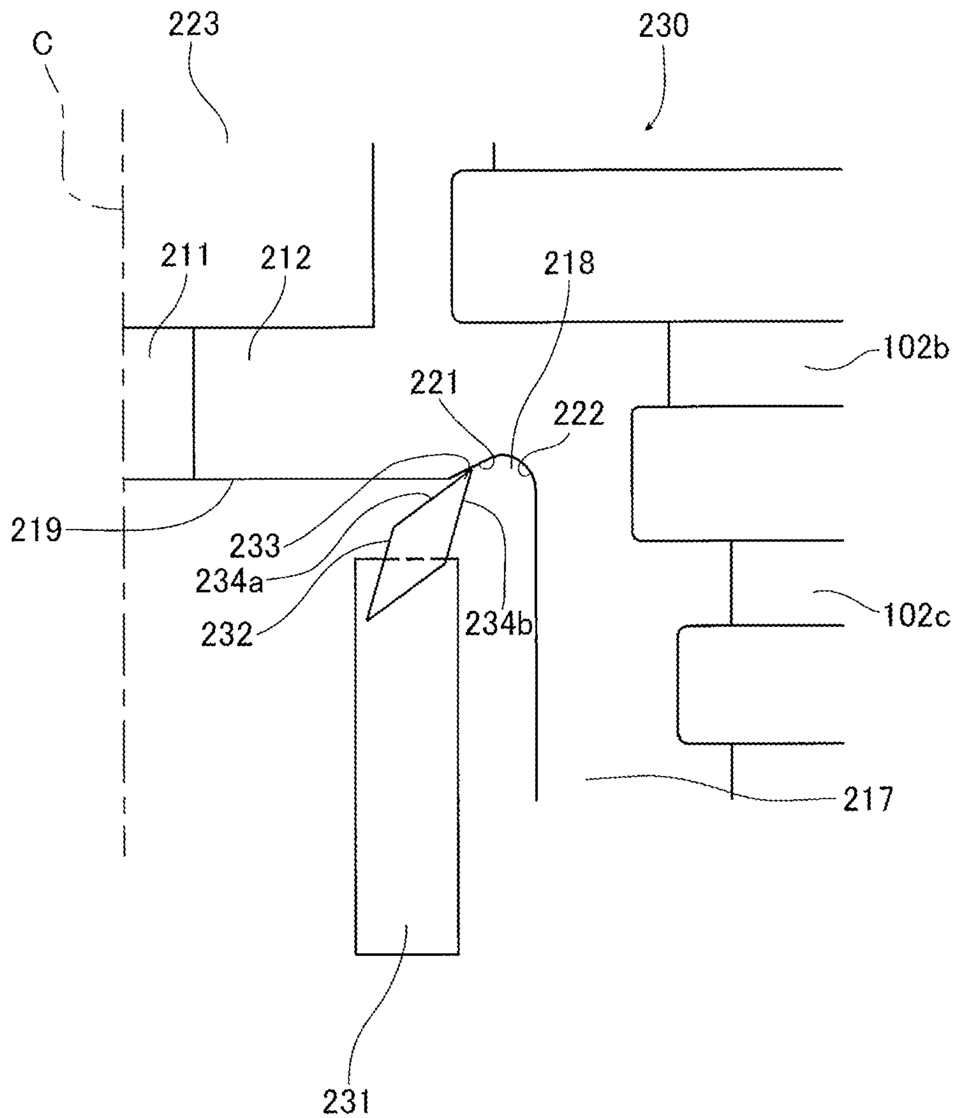
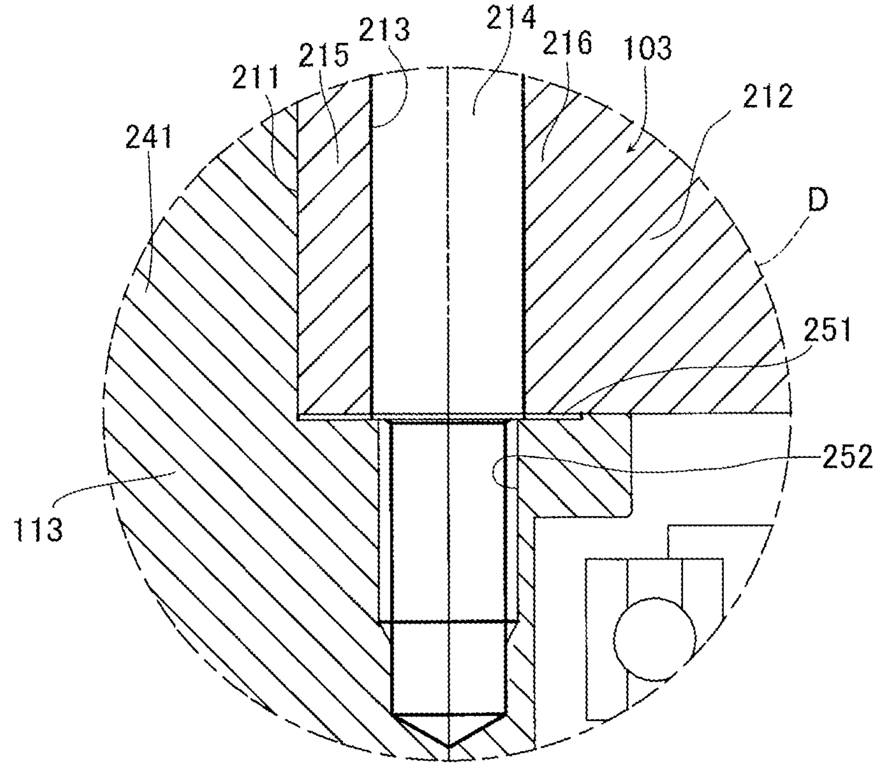
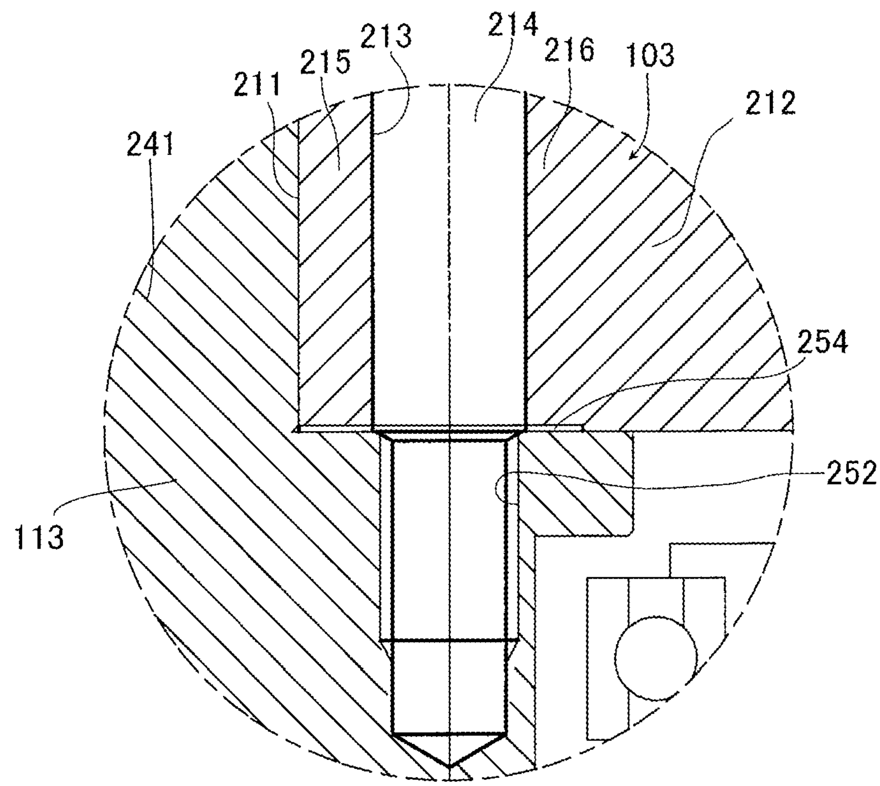

C
223
230
211
212
218
102b
221
222
219
233
234a
232
234b
102c
217
231

Fig.11



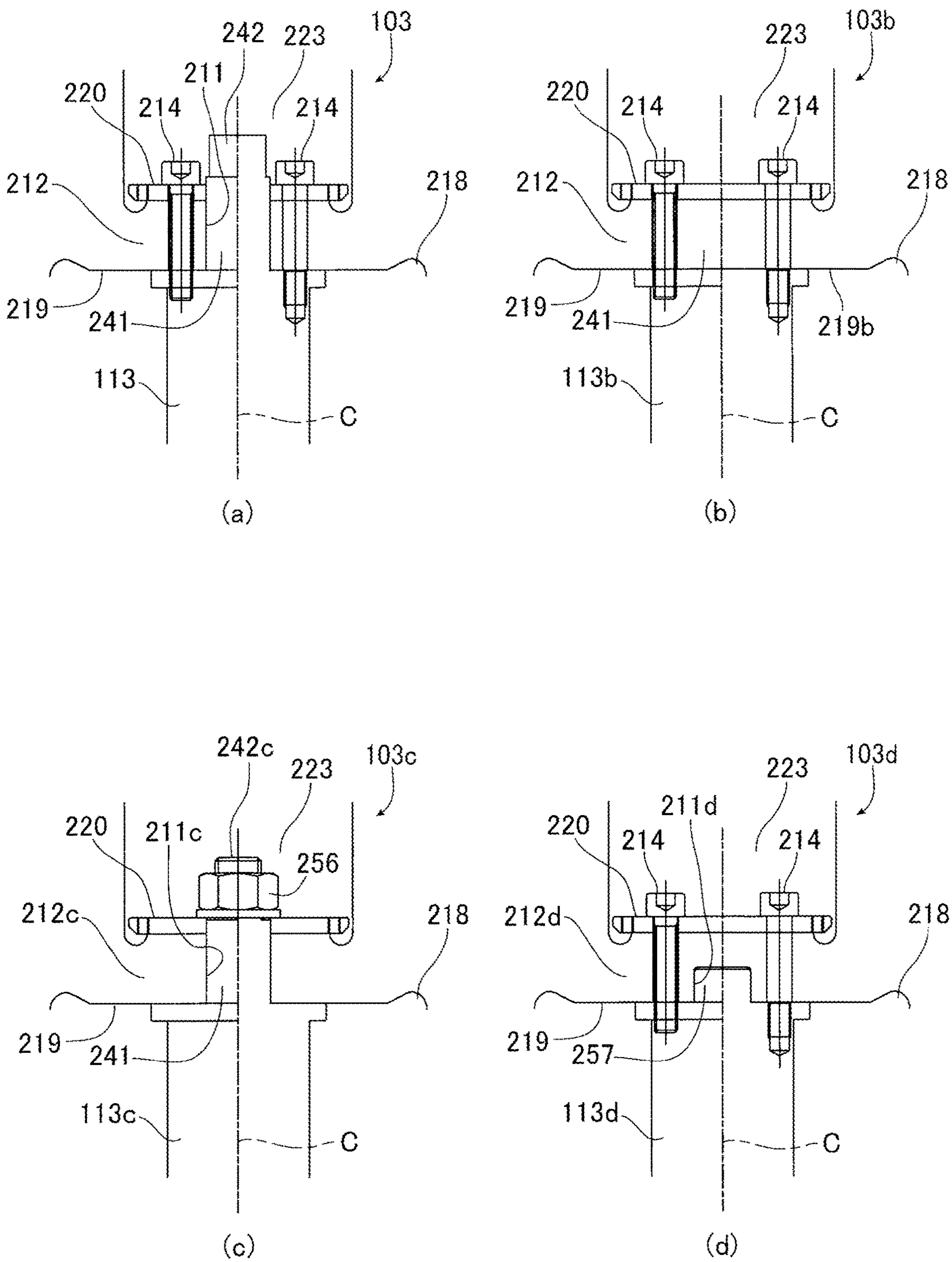
213
214
215
216
103
211
212
241
D
251
252
113

Fig.12



213
214
215
216
103
211
212
241
254
252
113

Fig.13


103
242
223
211
220
214
214
212
218
219
241
113
C
(a)
103b
223
220
214
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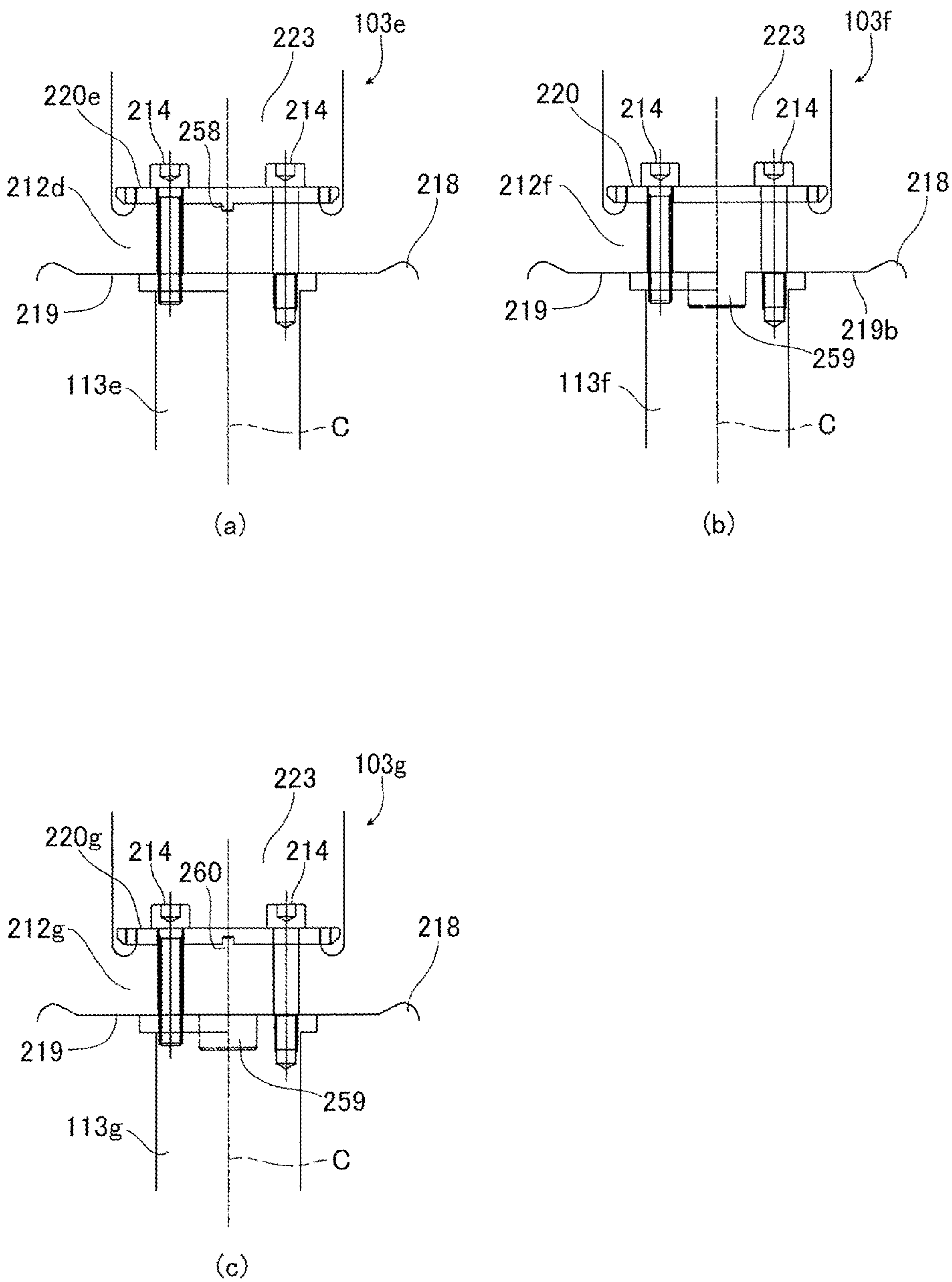
218
219
241
219b
113b
C
(b)
242c
223
103c
220
211c
256
212c
218
219
241
113c
C
(c)
103d
223
211d
220
214
214
212d
218
219
257
113d
C
(d)

Fig.14


103e
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220e
214
258
214
212d
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219
113e
C
(a)
103f
223
220
214
214
212f
218
219
219b
259
113f
C
(b)
103g
223
220g
214
260
214
212g
218
219
259
113g
C
(c)

VACUUM PUMP AND ROTATING BODY FOR VACUUM PUMP

CROSS-REFERENCE OF RELATED APPLICATION

This application is a Section 371 National Stage Application of International Application No. PCT/JP2022/005206, filed Feb. 9, 2022, which is incorporated by reference in its entirety and published as WO 2022/176745A1 on Aug. 25, 2022 and which claims priority of Japanese Application No. 2021-025072, filed Feb. 19, 2021.

FIELD

The present invention relates to a vacuum pump such as a turbo molecular pump and a rotating body for the vacuum pump, for example.

BACKGROUND

In general, a turbo molecular pump is known as one type of a vacuum pump. This turbo molecular pump is configured such that a rotor blade is rotated by conduction to a motor inside a pump main-body so as to flick off gas molecules of a gas (process gas) sucked into the pump main-body, whereby the gas is exhausted. Moreover, in this type of the turbo molecular pump, a rotating shaft (rotor shaft) is connected to the rotating body, on which the rotor blade is formed, so that the rotating shaft and the rotating body are rotated by the motor and perform exhaustion.

The discussion above is merely provided for general background information and is not intended to be used as an aid in determining the scope of the claimed subject matter. The claimed subject matter is not limited to implementations that solve any or all disadvantages noted in the background.

SUMMARY

By the way, in the vacuum pump such as various turbo molecular pumps as described above, in general the more a rotation number of the rotor blade is increased, the higher the exhaust performance becomes. However, as a result of structural analysis by the inventor and the like by paying attention to a connected part between the rotating body and the rotating shaft, it has been found out that stress concentration can easily occur at the connected part.

In order to reduce such stress generated at the connected part between the rotating body and the rotating shaft, setting of a rotation number at low during a rated operation can be considered, but if the rotation number is lowered, improvement of the exhaust performance becomes difficult. On the other hand, if the rotation number is increased in order to improve the exhaust performance without considering a measure against the stress concentration as above, a high stress is generated at the connected part between the rotating body and the rotating shaft, which lowers reliability. And since the connected part between the rotating body and the rotating shaft is a part which considerably influences reliability of the vacuum pump, a large design change of the connected part is not easy.

An object of the present invention is to provide a vacuum pump and a rotating body for the vacuum pump which can reduce stress concentration generated in the rotating body or particularly the stress generated at the connected part between the rotating body and the rotating shaft.

(1) In order to achieve the aforementioned object, the present invention is, in a vacuum pump having a rotor blade on an outer periphery of a cylinder part and including a rotating body fastened to a rotating shaft by a fastening means and rotatable together with the rotating shaft, characterized in that

in the rotating body, at least one of a fitting hole portion fitted with the rotating shaft and a through hole portion, through which the fastening means penetrates, is a stress-reduction target portion, and a groove portion that reduces stress generated in the stress-reduction target portion during rotation of the rotating body is provided in the rotating body.

(2) Moreover, in order to achieve the aforementioned object, another present invention is a vacuum pump described in (1), characterized in that the groove portion is provided closer to an outer peripheral side than the fitting hole portion or the through hole portion.

(3) Moreover, in order to achieve the aforementioned object, another present invention is a vacuum pump described in (1) or (2), characterized in that the groove portion is provided on at least one of an inner peripheral surface and an outer peripheral surface of the rotating body.

(4) Moreover, in order to achieve the aforementioned object, another present invention is a vacuum pump described in any one of (1) to (3), characterized in that, in the groove portion, at least an inner peripheral side has a gentle inclination structure.

(5) Moreover, in order to achieve the aforementioned object, another present invention is a vacuum pump described in any one of (1) to (4), characterized in that the groove portion is disposed on a fastened surface of the rotating body or on an extended surface thereof.

(6) Moreover, in order to achieve the aforementioned object, another present invention is a vacuum pump described in any one of (1) to (5), characterized in that the rotating body is applied with surface treatment and has a counterbore portion, which avoids contact with the rotating shaft, on a peripheral edge part of at least one of the fitting hole portion and the through hole portion.

(7) Moreover, in order to achieve the aforementioned object, another present invention is, in a rotating body for a vacuum pump having a rotor blade on an outer periphery of a cylinder part and fastened to the rotating shaft by a fastening means, characterized in that

at least one of a fitting hole portion fitted with the rotating shaft and a fastened part, to which the fastening means is fastened, is a stress-reduction target portion, the rotating body including a groove portion which reduces a stress generated in the stress-reduction target portion during rotation.

According to the present invention, the vacuum pump and the rotating body for the vacuum pump which can reduce stress generated in the connected part between the rotating body and the rotating shaft can be provided.

The Summary is provided to introduce a selection of concepts in a simplified form that are further described in the Detail Description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a vertical sectional view of a turbo molecular pump according to an embodiment of the present invention.

FIG. **2** is a circuit diagram of an amplifier circuit.

FIG. **3** is a time chart illustrating control when a current instruction value is larger than a detected value.

FIG. **4** is a time chart illustrating control when the current instruction value is smaller than the detected value.

FIG. **5** is a vertical sectional view illustrating a part of FIG. **1** in an enlarged manner.

FIG. **6** is a vertical sectional view illustrating a part of FIG. **5** in an enlarged manner.

FIG. **7** is an explanatory view illustrating a groove portion in the turbo molecular pump in FIG. **1** in an enlarged manner.

FIG. **8** is an explanatory view illustrating a variation of the groove portion.

FIG. **9**(*a*) is an explanatory diagram illustrating deflection of a rotating body when the groove portion is not provided, and FIG. **9**(*b*) is an explanatory diagram illustrating the deflection of the rotating body when the groove portion is provided.

FIG. **10** is an explanatory diagram schematically illustrating a state of a work when the groove portion is fabricated.

FIG. **11** is a vertical sectional view illustrating a stress-control recess portion formed in a part surrounded by a circle D in FIG. **1** in an enlarged manner.

FIG. **12** is a vertical sectional view illustrating a variation of the stress-control recess portion.

FIG. **13**(*a*) is an explanatory diagram illustrating a connection relationship between the rotating body and a rotor shaft shown in FIG. **1** and FIG. **5**, FIG. **13**(*b*) is an explanatory diagram illustrating a variation of the rotating body, FIG. **13**(*c*) is an explanatory diagram illustrating another variation of the rotating body, and FIG. **13**(*d*) is an explanatory diagram illustrating still another variation of the rotating body.

FIG. **14**(*a*) is an explanatory diagram illustrating still another variation of the rotating body, FIG. **14**(*b*) is an explanatory diagram illustrating still another variation of the rotating body, and FIG. **14**(*c*) is an explanatory diagram illustrating still another variation of the rotating body.

DETAILED DESCRIPTION

Hereinafter, a vacuum pump according to an embodiment of the present invention will be explained on the basis of the drawings. FIG. **1** illustrates a turbo molecular pump **100** as a vacuum pump according to the embodiment of the present invention. This turbo molecular pump **100** is configured to be connected to a vacuum chamber (not shown) of a target device such as a semiconductor manufacturing device, for example.

A vertical sectional view of this turbo molecular pump **100** is shown in FIG. **1**. In FIG. **1**, the turbo molecular pump **100** has an inlet port **101** formed in an upper end of a cylindrical outer cylinder **127**. And inside the outer cylinder **127**, a rotating body **103** with a plurality of rotor blades **102** (**102***a*, **102***b*, **102***c*, . . .), which are turbine blades for sucking/exhausting a gas, formed on a peripheral part radially and in multiple stages is provided. At a center of this rotating body **103**, a rotor shaft **113** (rotating shaft) is mounted, and this rotor shaft **113** is floated/supported in the air and position-controlled by a magnetic bearing of 5-axis control, for example. The rotating body **103** is constituted by metal such as aluminum or an aluminum alloy in general.

Regarding an upper-side radial electromagnet **104**, four electromagnets are disposed by forming a pair on an X-axis and a Y-axis. Four upper-side radial sensors **107** are provided in the vicinity of the upper-side radial electromagnets **104** and correspondingly to each of the upper-side radial electromagnets **104**. As the upper-side radial sensors **107**, an inductance sensor, an eddy current sensor or the like having a conductive winding is used, and a position of the rotor shaft **113** is detected on the basis of a change in inductance of this conductive winding changing in accordance with the position of the rotor shaft **113**. This upper-side radial sensor **107** is configured to detect radial displacement of the rotor shaft **113**, that is, the rotating body **103** fixed thereto and to send it to a control device **200**.

In this control device **200**, a compensation circuit having a PID adjustment function, for example, generates an excitation control-instruction signal of the upper-side radial electromagnet **104** on the basis of a position signal detected by the upper-side radial sensor **107**, and an amplifier circuit **150** (which will be described later) shown in FIG. **2** excites and controls the upper-side radial electromagnet **104** on the basis of this excitation control-instruction signal, whereby an upper side radial position of the rotor shaft **113** is adjusted.

And this rotor shaft **113** is formed of a material with high magnetic permeability (such as iron and stainless) and the like and is attracted by a magnetic force of the upper-side radial electromagnet **104**. Such adjustment is performed independently in each of an X-axis direction and a Y-axis direction. Moreover, a lower-side radial electromagnet **105** and a lower-side radial sensor **108** are disposed similarly to the upper-side radial electromagnet **104** and the upper-side radial sensor **107** and adjust a radial position on a lower side of the rotor shaft **113** similarly to the radial position on the upper side.

Moreover, axial electromagnets **106**A, **106**B are disposed by vertically sandwiching a disc-shaped metal disc **111** provided on a lower part of the rotor shaft **113**. The metal disc **111** is constituted by a material with high magnetic permeability such as iron. In order to detect axial displacement of the rotor shaft **113**, the axial sensor **109** is provided, and it is configured such that an axial position signal thereof is sent to the control device **200**.

And in the control device **200**, the compensation circuit having the PID adjustment function, for example, generates an excitation control-instruction signal for each of the axial electromagnet **106**A and the axial electromagnet **106**B on the basis of the axial position signal detected by the axial sensor **109**, and the amplifier circuit **150** excites and controls the axial electromagnet **106**A and the axial electromagnet **106**B, respectively, on the basis of these excitation control-instruction signals, whereby the axial electromagnet **106**A attracts the metal disc **111** upward by the magnetic force, while the axial electromagnet **106**B attracts the metal disc **111** downward, and the axial position of the rotor shaft **113** is adjusted.

As described above, the control device **200** adjusts the magnetic force by which the axial electromagnets **106**A, **106**B affect the metal disc **111** as appropriate, magnetically floats the rotor shaft **113** in the axial direction, and holds it in a space in a non-contact manner. Note that, the amplifier circuit **150** which excites and controls the upper-side radial electromagnet **104**, the lower-side radial electromagnet **105**, and the axial electromagnets **106**A, **106**B will be described later.

On the other hand, a motor **121** includes a plurality of magnetic poles disposed in a peripheral state so as to surround the rotor shaft **113**. Each of the magnetic poles is controlled by the control device **200** so as to rotate and drive the rotor shaft **113** via an electromagnetic force acting between it and the rotor shaft **113**. Moreover, rotation speed sensors such as a Hall element, a resolver, and an encoder, not shown, for example, are incorporated in the motor **121** such that a rotation speed of the rotor shaft **113** is detected by a detection signal of this rotation speed sensor.

Furthermore, a phase sensor, not shown, is mounted in the vicinity of the lower-side radial sensor **108**, for example, so as to detect a phase of rotation of the rotor shaft **113**. The control device **200** is configured to detect a position of the magnetic pole by using the detection signals of this phase sensor and the rotation speed sensor together.

A plurality of stator blades **123** (**123***a*, **123***b*, **123***c*, . . .) are disposed with a slight gap from the rotor blades **102** (**102***a*, **102***b*, **102***c*, . . .). The rotor blades **102** (**102***a*, **102***b*, **102***c*, . . .) are formed with inclination only by a predetermined angle from a plane perpendicular to an axis of the rotor shaft **113** in order to transfer molecules of an exhaust gas downward by collision, respectively. The stator blades **123** (**123***a*, **123***b*, **123***c*, . . .) are constituted by metal such as aluminum, iron, stainless, copper, for example, or an alloy containing these metals as components.

Moreover, the stator blades **123** are also formed with inclination only by a predetermined angle from a plane perpendicular to the axis of the rotor shaft **113** similarly and are disposed alternately with stages of the rotor blades **102** toward the inside of the outer cylinder **127**. And an outer peripheral end of the stator blade **123** is supported in a state fitted and inserted between a plurality of stator-blade spacers **125** (**125***a*, **125***b*, **125***c*, . . .) stacked in stages.

The stator-blade spacer **125** is a ring-shaped member and is constituted by metal such as aluminum, iron, stainless, copper, for example, or an alloy containing these metals as components. On an outer periphery of the stator-blade spacer **125**, the outer cylinder **127** is fixed with a slight gap. A base portion **129** is disposed on a bottom part of the outer cylinder **127**. An outlet port **133** is formed in the base portion **129** and is made to communicate with the outside. The exhaust gas entering the inlet port **101** from the chamber (vacuum chamber) side and transferred to the base portion **129** is sent to the outlet port **133**.

Moreover, depending on an application of the turbo molecular pump **100**, the threaded spacer **131** is disposed between the lower part of the stator-blade spacer **125** and the base portion **129**. The threaded spacer **131** is a cylindrical member constituted by metal such as aluminum, copper, stainless, iron or an alloy having these metals as components, and a plurality of spiral thread grooves **131***a* are engraved in an inner peripheral surface thereof. A spiral direction of the thread groove **131***a* is a direction in which, when molecules of the exhaust gas move in a rotating direction of the rotating body **103**, the molecules are transferred toward the outlet port **133**. At a lowest part continuing to the rotor blades **102** (**102***a*, **102***b*, **102***c*, . . .) of the rotating body **103**, a cylinder portion **102***d* is suspended. An outer peripheral surface of this cylinder portion **102***d* is cylindrical and is extended toward the inner peripheral surface of the threaded spacer **131** and is in the vicinity of the inner peripheral surface of this threaded spacer **131** with a predetermined gap. The exhaust gas having been transferred to the thread groove **131***a* by the rotor blade **102** and the stator blade **123** is sent to the base portion **129** while being guided by the thread groove **131***a*.

The base portion **129** is a disc-shaped member constituting a base part of the turbo molecular pump **100** and is constituted by metal in general such as iron, aluminum, and stainless. The base portion **129** physically holds the turbo molecular pump **100** and also functions as a heat conduction path and thus, metal with rigidity and high heat conductivity such as iron, aluminum, and copper is preferably used.

In the configuration as above, when the rotor blade **102** is rotated and driven by the motor **121** together with the rotor shaft **113**, the exhaust gas is sucked from the chamber through the inlet port **101** by actions of the rotor blade **102** and the stator blade **123**. A rotation speed of the rotor blade **102** is normally 20000 rpm to 90000 rpm, and a peripheral speed at a distal end of the rotor blade **102** reaches 200 m/s to 400 m/s. The exhaust gas sucked from the inlet port **101** passes between the rotor blade **102** and the stator blade **123** and is transferred to the base portion **129**. At this time, a friction heat generated when the exhaust gas is brought into contact with the rotor blade **102** and conduction of the heat generated in the motor **121** raise a temperature of the rotor blade **102**, and this heat is transmitted by radiation or conduction by gas molecules of the exhaust gas and the like to the stator blade **123** side.

The stator-blade spacers **125** are joined to each other on the outer peripheral parts, and the heat received by the stator blade **123** from the rotor blade **102** or the friction heat generated when the exhaust gas contacts the stator blade **123** or the like is transmitted to the outside.

Note that, in the explanation described above, the threaded spacer **131** is disposed on the outer periphery of the cylinder portion **102***d* of the rotating body **103**, and the thread groove **131***a* is engraved in the inner peripheral surface of the threaded spacer **131**. However, to the contrary, the thread groove is engraved in the outer peripheral surface of the cylinder portion **102***d*, and the spacer having a cylindrical inner peripheral surface is disposed in the periphery thereof in some cases.

Moreover, depending on the application of the turbo molecular pump **100**, an electric component portion is covered with a stator column **122** on the periphery, and inside this stator column **122** is kept at a predetermined pressure by a purge gas in some cases so that the gas sucked from the inlet port **101** does not intrude into the electric component portion constituted by the upper-side radial electromagnet **104**, the upper-side radial sensor **107**, the motor **121**, the lower-side radial electromagnet **105**, the lower-side radial sensor **108**, the axial electromagnets **106**A, **106**B, the axial sensor **109** and the like.

In this case, a piping, not shown, is disposed in the base portion **129**, and the purge gas is introduced through this piping. The introduced purge gas is sent out to the outlet port **133** through gaps between a protective bearing **120** and the rotor shaft **113**, between a rotor and a stator of the motor **121**, and between the stator column **122** and the inner-peripheral side cylinder portion of the rotor blade **102**.

Here, the turbo molecular pump **100** requires control based on specification of a model and individually adjusted specific parameters (characteristics corresponding to the model, for example). In order to store the control parameters, the turbo molecular pump **100** includes an electronic circuit portion **141** inside the main body thereof. The electronic circuit portion **141** is constituted by a semiconductor memory such as EEP-ROM, and electronic components such as semiconductor elements for access thereof, a board **143** for mounting them and the like. This electronic circuit portion **141** is accommodated in a lower part of a rotation speed sensor, not shown, in the vicinity of a center of the base portion **129**, for example, constituting the lower part of the turbo molecular pump **100** and is closed by an airtight bottom lid **145**.

By the way, in a manufacturing process of a semiconductor, some of process gases introduced into the chamber have such a nature that becomes a solid when its pressure becomes higher than a predetermined value or when its temperature becomes lower than a predetermined value. Inside the turbo molecular pump **100**, a pressure of the exhaust gas is the lowest at the inlet port **101** and the highest at the outlet port **133**. If the pressure of the process gas becomes higher than the predetermined value or the temperature thereof becomes lower than the predetermined value in the middle of transfer from the inlet port **101** to the outlet port **133**, the process gas becomes a solid state, which adheres and deposits inside the turbo molecular pump **100**.

When SiCl4 is used as the process gas in an Al etching device, for example, at a low vacuum (760 [torr] to 10-2 [torr]) and at a low temperature (approximately 20 KA it is known from a steam pressure curve that a solid product (AlCl3, for example) precipitates, adheres and deposits inside the turbo molecular pump **100**. As a result, if the precipitates of the process gas deposit inside the turbo molecular pump **100**, the depositions narrow a pump channel and causes lowering of performances of the turbo molecular pump **100**. And the product described above was in such a state that easily solidifies and adheres at a part in the vicinity of the outlet port **133** or in the vicinity of the threaded spacer **131** where the pressure is high.

Therefore, in order to solve this problem, conventionally, a heater or an annular water-cooling pipe **149**, not shown, is wrapped around an outer periphery of the base portion **129** and the like, a temperature sensor (thermistor, for example), not shown, is embedded in the base portion **129**, for example, and control of heating of the heater or cooling by the water-cooling pipe **149** is executed so as to keep the temperature of the base portion **129** at a certain high temperature (set temperature) on the basis of a signal of this temperature sensor (hereinafter, called TMS. TMS: Temperature Management System).

Subsequently, regarding the turbo molecular pump **100** configured as above, the amplifier circuit **150** that excites and controls the upper-side radial electromagnet **104**, the lower-side radial electromagnet **105**, and the axial electromagnets **106**A, **106**B will be explained. A circuit diagram of this amplifier circuit **150** is shown in FIG. **2**.

In FIG. **2**, an electromagnet winding **151** constituting the upper-side radial electromagnet **104** and the like has one end thereof connected to a positive pole **171***a* of a power source **171** through a transistor **161** and the other end connected to a negative pole **171***b* of the power source **171** through a current detection circuit **181** and a transistor **162**. And the transistors **161**, **162** are so-called power MOSFET and have a structure in which a diode is connected between source-drain thereof.

At this time, the transistor **161** has a cathode terminal **161***a* of the diode thereof connected to the positive pole **171***a* and an anode terminal **161***b* connected to one end of the electromagnet winding **151**. Moreover, the transistor **162** has a cathode terminal **162***a* of the diode thereof connected to the current detection circuit **181** and an anode terminal **162***b* connected to the negative pole **171***b*.

On the other hand, a diode **165** for current regeneration has a cathode terminal **165***a* thereof connected to one end of the electromagnet winding **151** and an anode terminal **165***b* thereof connected to the negative pole **171***b*. Moreover, similarly, a diode **166** for current regeneration has a cathode terminal **166***a* thereof connected to the positive pole **171***a* and an anode terminal **166***b* thereof connected to the other end of the electromagnet winding **151** through the current detection circuit **181**. And the current detection circuit **181** is constituted by a Hall sensor-type current sensor or an electric resistance element, for example.

The amplifier circuit **150** constituted as above corresponds to one electromagnet. Thus, in a case where the magnetic bearing is 5-axis control and there are ten pieces in total of the electromagnets **104**, **105**, **106**A, **106**B, the similar amplifier circuit **150** is constituted for each of the electromagnets, and ten units of the amplifier circuits **150** are connected to the power source **171** in parallel.

Moreover, an amplifier control circuit **191** is constituted by a digital-signal processing portion (hereinafter referred to as a DSP portion), not shown, of the control device **200**, for example, and this amplifier control circuit **191** switches on/off of the transistors **161**, **162**.

The amplifier control circuit **191** compares a current value detected by the current detection circuit **181** (a signal reflecting this current value is called a current detection signal **191***c*) with a predetermined current instruction value. And on the basis of this comparison result, a size of a pulse width (pulse width time Tp**1**, Tp**2**) generated within a control cycle Ts, which is one cycle by PWM control, is determined. As a result, gate drive signals **191***a*, **191***b* having this pulse width are output from the amplifier control circuit **191** to gate terminals of the transistors **161**, **162**.

Note that, when passing a resonant point during an acceleration operation of a rotation speed of the rotating body **103** or when disturbance occurs during a constant-speed operation and the like, position control of the rotating body **103** needs to be performed at a high speed and with a strong force. Thus, a high voltage such as 50V, for example, is used for the power source **171** so that the current flowing in the electromagnet winding **151** can rapidly increase (or decrease). Moreover, between the positive pole **171***a* and the negative pole **171***b* of the power source **171**, an ordinary capacitor is connected (not shown) for stabilization of the power source **171**.

In the configuration as above, when both the transistors **161**, **162** are turned on, the current flowing through the electromagnet winding **151** (hereinafter, referred to as an electromagnet current iL) increases, while when the both are turned off, the electromagnet current iL decreases.

Moreover, when one of the transistors **161**, **162** is turned on, while the other is turned off, a so-called flywheel current is held. And by causing the flywheel current to flow through the amplifier circuit **150** as above, a hysteresis loss in the amplifier circuit **150** is decreased, and power consumption of the circuit as a whole can be kept low. Moreover, by controlling the transistors **161**, **162** as above, a high frequency noise such as a higher harmonic wave generated in the turbo molecular pump **100** can be reduced. Furthermore, by measuring this flywheel current by the current detection circuit **181**, the electromagnet current iL flowing through the electromagnet winding **151** can be detected.

That is, if the detected current value is smaller than the current instruction value, both the transistors **161**, **162** are turned on only for a period of time corresponding to the pulse width time Tp**1** only once in the control cycle Ts (100 μs, for example) as shown in FIG. **3**. Thus, the electromagnet current iL during this period increases toward a current value iLmax (not shown) that can flow from the positive pole **171***a* toward the negative pole **171***b* through the transistors **161**, **162**.

On the other hand, if the detected current value is larger than the current instruction value, both the transistors **161**, **162** are turned off only for a period of time corresponding to the pulse width time Tp**2** only once in the control cycle Ts as shown in FIG. **4**. Thus, the electromagnet current iL during this period decreases toward a current value iLmin (not shown) that can be regenerated from the negative pole **171***b* toward the positive pole **171***a* through the diodes **165**, **166**.

And in the both cases, after elapse of the pulse width time Tp**1**, Tp**2**, either one of the transistors **161**, **162** is turned on. Thus, during this period, the flywheel current is held in the amplifier circuit **150**.

In the turbo molecular pump **100** having the basic configuration as above, an upper side in FIG. **1** (the side of the inlet port **101**) is a sucking portion connecting to the side of the target device, and a lower side (the side provided on the base portion **129** so that the outlet port **133** protrudes to the left side in the drawing) side is an exhaust portion connecting to an auxiliary pump (back pump for roughing), not shown, and the like. And the turbo molecular pump **100** can be used in an inverted attitude, a horizontal attitude, an inclined attitude other than a perpendicular attitude in a vertical direction as shown in FIG. **1**.

Moreover, in the turbo molecular pump **100**, the outer cylinder **127** described above and the base portion **129** are combined to constitute a single case (hereinafter, both are combined and called a "main-body casing" and the like in some cases). Moreover, the turbo molecular pump **100** is connected to a box-shaped electric component case (not shown) electrically (and structurally), and the electric component case incorporates the control device **200** described above.

An internal constitution of the main-body casing (combination of the outer cylinder **127** and the base portion **129**) of the turbo molecular pump **100** can be separated into a rotation mechanism portion that rotates the rotor shaft **113** and the like by the motor **121** and an exhaust mechanism portion rotated/driven by the rotation mechanism portion. Moreover, the exhaust mechanism portion can be considered separately as a turbo-molecular-pump mechanism portion constituted by the rotor blade **102**, the stator blade **123** and the like and a groove-exhaust mechanism portion (which will be described later) constituted by the cylinder portion **102***d*, the threaded spacer **131** and the like.

Moreover, the purge gas (protective gas) described above is used for protection of the bearing part, the rotor blade **102** and the like and prevents corrosion caused by the exhaust gas (process gas) and cools the rotor blade **102** and the like. Supply of this purge gas can be performed by a general method.

For example, though not shown, a purge gas channel extending linearly in the radial direction is provided at a predetermined part (a position separated approximately by 180 degrees with respect to the outlet port **133** and the like) of the base portion **129**. And to this purge gas channel (more specifically, a purge port to be an inlet of the gas), a purge gas is supplied via a purge-gas bomb (N2 gas bomb or the like), a flow-rate controller (valve device) and the like from the outer side of the base portion **129**.

The aforementioned protective bearing **120** is also called a "touchdown (T/D) bearing", "backup bearing" and the like. By means of these protective bearings **120**, even in the case of a trouble in an electric system or a trouble such as atmospheric entry or the like, for example, a position or an attitude of the rotor shaft **113** is not largely changed, or the rotor blade **102** or its peripheral part is not damaged.

Note that, in each of the drawings (FIG. **1**, FIGS. **5** to **10**, FIG. **13**, FIG. **14**) showing the structure of the turbo molecular pump **100**, depiction of hatching illustrating a section of a component is omitted in order to avoid complexity of the drawings.

Subsequently, a stress distribution function and the like of the rotating body **103** described above will be explained. As described above, the rotor shaft **113** is mounted at the center of the rotating body **103**. The rotating body **103** has a disc portion **212** having a fitting hole **211** at the center as shown in FIG. **5** in an enlarged manner, and the fitting shaft portion **241** of the rotor shaft **113** is fitted in a fitting hole **211**.

On one end part in the axial direction of the rotor shaft **113** (upper end part in FIG. **1** and FIG. **5**, here), a relatively small-diameter protruding end portion **242** and the aforementioned fitting shaft portion **241** are formed. The protruding end portion **242** and the fitting shaft portion **241** are formed with diameters different from each other, and the diameter of the fitting shaft portion **241** is larger than the diameter of the protruding end portion **242**.

Furthermore, the fitting shaft portion **241** and the protruding end portion **242** constitute a stepped shape. And the fitting shaft portion **241** is coaxially inserted into the fitting hole **211** of the rotating body **103** and is brought into contact with an inner peripheral surface of the fitting hole **211** with a pressure generated by a predetermined method (here, quenching). Though not shown clearly, a length in the axial direction acting as the fitting part of the fitting shaft portion **241** substantially matches a thickness H of the disc portion **212** shown in FIG. **5**. Moreover, the protruding end portion **242** of the rotor shaft **113** is located outside the disc portion **212** of the rotating body **103**.

In the disc portion **212** of the rotating body **103**, a plurality of (6 spots or 8 spots, for example) bolt through holes **213** are formed and disposed in the periphery of the fitting hole **211**. In the bolt through holes **213**, a bolt **214** (fastening means) such as a bolt with a hexagonal hole or the like is inserted. These bolts **214** are threaded into the rotor shaft **113**. And the rotating body **103** and the rotor shaft **113** are coupled with each other by a fastening force of the bolt **214**.

In the following, as shown in FIG. **6**, the fitting hole **211** and the parts around the fitting hole **211** in the disc portion **212** of the rotating body **103** are assumed to be a fitting hole portion **215** (stress-reduction target portion). Moreover, the bolt through hole **213** and the part around the bolt through hole **213** in the disc portion **212** are assumed to be a through hole portion **216** (similarly, the stress-reduction target portion). Moreover, when the fitting hole portion **215** and the through hole portion **216** are adjacent to each other, an area where the fitting hole portion **215**, which is the stress-reduction target portion, and the through hole portion **216**, which is the stress-reduction target portion, overlap each other is assumed to be generated. And the overlap area in this case is also assumed to be the stress-reduction target portion. Here, in this embodiment, the explanation is made such that the fitting hole portion **215** includes the fitting hole **211**, and the through hole portion **216** includes the bolt through hole **213**, but this is not limiting, and it may be configured such that the fitting hole portion **215** does not include the fitting hole **211**, and the through hole portion **216** does not include the bolt through hole **213**.

Moreover, between the disc portion **212** and the head part of the bolt **214**, a disc-shaped and annular washer **220** is sandwiched. Furthermore, on an outer peripheral part on the bottom part in the recessed portion **223** in which the washer **220** is disposed in the disc portion **212**, as shown in FIG. **6**, a groove portion (groove portion in recessed portion) **223***a* facing a plate surface of the washer **220** is formed. Moreover, though reference numerals are omitted, a plurality of through holes capable of preventing collecting of a gas between the disc portion **212** and itself is formed in the washer **220**.

On the outer peripheral part of the disc portion **212** in the rotating body **103**, as shown in FIG. **5**, a rotor-blade forming portion **217** (cylinder part) is integrally formed continuously. This rotor-blade forming portion **217** is integrally formed continuously also on the cylinder portion **102***d* described above of the rotating body **103**. Moreover, at a boundary part **224** between disc portion **212** and the rotor-blade forming portion **217**, a groove portion **218** is formed.

This groove portion **218** is formed in an inner peripheral surface **219** of the disc portion **212** and is opened in the inner peripheral surface **219**. Moreover, in this embodiment, the groove portion **218** extends over the entire circumference of the inner peripheral surface **219** and has a constant sectional shape over the entire circumference. Note that the invention according to this embodiment is not limited to the formation of the groove portion **218** over the entire circumference of the inner peripheral surface **219**, but the groove portion **218** may be formed so as to be intermittently disposed along the peripheral direction of the inner peripheral surface **219**.

The groove portion **218** has, as shown in FIG. **7**, an inclined portion **221** and a curved portion **222**. The inclined portion **221** in them is inclined so as to be deeper from a center side toward an outer peripheral side of the rotating body **103**, and the curved portion **222** is curved in an arc shape so as to become shallow from the center side toward the outer peripheral side of the rotating body **103**. Furthermore, the inclined portion **221** is positioned on the center side of the rotating body **103**, and the curved portion **222** is positioned on the outer peripheral side of the inclined portion **221**.

The inclined portion **221** is formed at a part adjacent to the inner peripheral surface **219** of the disc portion **212** and continues from the inner peripheral surface **219** via an inclination angle $\alpha 1$ with the inner peripheral surface **219** as a reference. Moreover, the inclination angle $\alpha 1$ is substantially constant from the inner peripheral side to the outer peripheral side of the inclined portion **221**. Note that this is not limiting, and the inclination angle of the inclined portion **221** may be configured to be changed in the middle from the inner peripheral side toward the outer peripheral side.

The curved portion **222** is formed so as to have a tangent angle of $\alpha 2$ with the inner peripheral surface **219** of the disc portion **212** as a reference. In FIG. **7**, the tangent angle $\alpha 2$ is an angle of a tangent at a position where the curved portion **222** intersects an extension **219***a* of the inner peripheral surface **219**. Moreover, the inclination angle $\alpha 1$ and the tangent angle $\alpha 2$ have a relationship of $\alpha 1<\alpha 2$. That is, in a relationship between the tangent angle $\alpha 2$ and the inclination angle $\alpha 1$ in the groove portion **218**, the inclination angle $\alpha 1$ has a gentler inclination structure as compared with the tangent angle $\alpha 2$, and the tangent angle $\alpha 2$ has a larger and steeper inclination structure than the inclination angle $\alpha 1$.

Moreover, the inclination angle $\alpha 1$ preferably has an acute-angle structure ($\alpha<45$ degrees).

In the turbo molecular pump **100** having the rotating body **103** as above, the more the rotation number of the rotating body **103** is raised, the higher the exhaust performance becomes. However, when the rotating body **103** is designed, a shape and a dimension of each part need to be determined so that an excessive stress is not generated by a centrifugal force during rotation.

Moreover, spots where the stress concentration can easily occur in the rotating body **103** include the connected part between the rotating body **103** and the rotor shaft **113**. And the connected part between the rotating body **103** and the rotor shaft **113** includes the through hole portion **216** around the bolt through hole **213** and the fitting hole portion **215** around the fitting hole **211**.

If the stress generated in these spots can be reduced and an excessive rise of the stress generated in other spots can be prevented, strength of the rotating body **103** and the reliability of the turbo molecular pump **100** are improved. Moreover, since a room for the stress generation can be increased, the rotation number of the rotating body **103** can be raised, and the exhaust performance of the turbo molecular pump **100** can be improved.

From these viewpoints, the inventor and the like have made researches on the stress reduction in the through hole portion **216** and the fitting hole portion **215** and acquired an idea of purposely adding an irregular part to the disc portion **212** in the rotating body **103** so as to increase the stress generated in the disc portion **212**. And by forming the groove portion **218** as described above as the irregular part, the stress generated in the parts in the vicinity of the through hole portion **216** and the fitting hole portion **215** can be increased. As a result, the stress can be distributed in the rotating body **103** (particularly in the disc portion **212**), the stress in the through hole portion **216** and the fitting hole portion **215** can be reduced, and reliability and performances of the turbo molecular pump **100** can be improved.

Moreover, the inventor and the like conducted simulations for a structural model in which the groove portion **218** is formed in the rotating body **103** and experiments using an actual one, the stress generated in the through hole portion **216** and the fitting hole portion **215** was actually lowered. It can be considered that the groove portion **218** forms a so-called "escape" of the stress, and the stress was averaged.

Moreover, even if the stress generated in the through hole portion **216** and the fitting hole portion **215** can be reduced as described above, an excessive stress in the groove portion **218** is not preferable. Furthermore, excessive increases in the number of processes and costs for machining the rotating body **103** by providing the groove portion **218** are not preferable, either. Thus, the inventor and the like examined the optimal shape and machining method for the groove portion **218** and reached the conclusion that such a shape is preferable that the inclined portion **221** that inclines gently is disposed on the inner peripheral side and that a relatively steep part (the curved portion **222**, here) is disposed on the outer peripheral side. By forming the groove portion **218** having the shape as above on the rotating body **103**, the stress distribution can be distributed more averagely.

FIG. **9**(*a*) models and schematically illustrates deformation at rotation of the rotating body **103** in the case where the groove portion **218** is not provided (conventional structure). Moreover, FIG. **9**(*b*) similarly models and illustrates the deformation at the rotation of the rotating body **103** in the case where the groove portion **218** is provided. At the rotation of the rotating body **103**, a centrifugal force acts on the rotating body **103**, and a load F acts on the rotor-blade forming portion **217** to the outer peripheral side. Moreover, on the connected part between the rotating body **103** and the rotor shaft, a moment by the load F acts.

And the rotating body **103** is deformed with the fastened part with the rotor shaft **113** (here, it is considered as the through hole portion **216**) as a fulcrum, and the closer it gets from the fulcrum toward the outer peripheral side, the larger it is deflected and displaced to the sucking side (Upper sides in FIGS. **9**(*a*), **9**(*b*)). In FIGS. **9**(*a*), **9**(*b*), it is assumed that displacement amounts in the axial direction on the outer peripheral side of the disc portion **212** are δa, δb, respectively, and displacement amounts in the radial direction on the exhaust side in the rotor-blade forming portion **217** (lower sides in FIGS. **9**(*a*), **9**(*b*)) are γa, γb.

When the groove portion **218** is provided as shown in FIG. **9**(*b*), in the displacement amounts δa, δb of the disc portion **212**, δb becomes smaller than δa for the reason described below. In the displacement amounts γa, γb of the rotor-blade forming portion **217** whose lower side is not restricted, γb becomes larger than γa. That is, in the case of FIG. **9**(*b*) in which the groove portion **218** is provided, the stress generated in the vicinity of the groove portion **218** becomes higher than the case of FIG. **9**(*a*) in which the groove portion **218** is not provided, and the deformation of the rotor-blade forming portion **217** becomes larger on the outer peripheral side than the groove portion **218**. And the stress corresponding to a part by which the deformation amount increases is generated on and around the groove portion **218**, and the stress caused by the load F is distributed not only to the fastened part (fulcrum) but also to the groove portion **218**.

And by distributing the stress caused by the load F to the groove portion **218**, the stress generated at the fastened part (fulcrum) can be reduced. And by keeping the stress generated in the groove portion **218** appropriate and by preventing the excessive rise of the stress generated in the groove portion **218**, the strength of the rotating body **103** and the reliability of the turbo molecular pump **100** are improved as a whole. Moreover, since a room for stress generation can be increased, the rotation number of the rotating body **103** can be raised, and the exhaust performance of the turbo molecular pump **100** can be improved.

Subsequently, when the groove portion **218** as above is to be fabricated, a work as shown in FIG. **10** can be performed. For example, while a base material **230** of the rotating body **103** is rotated around a shaft center, a cutting tool (cutting tool) **231** is made to advance to the inner peripheral side of the base material **230**. Here, what is indicated by a reference character C in FIG. **10** is the shaft center of the base material **230**, and at the fabrication of the groove portion **218**, the base material **230** is rotated around this shaft center C. In FIG. **10**, only a part of the half in the base material **230** is shown with the shaft center C as a boundary.

At a distal end of the cutting tool **231**, a tip (blade edge) **232** for lathe is mounted. At the distal end of the tip **232**, cutting-edge surfaces **234***a*, **234***b* are formed by sandwiching an angle portion **233**. The tip **232** is mounted on the cutting tool **231** with the distal end side directed to the outer peripheral side (outer side in the radial direction) of the disc portion **212** in the base material **230**.

The cutting tool **231** performs forward/backward movement along the shaft center C of the base material **230** or vertical/lateral movement on an orthogonal plane to the shaft center C by a feeding mechanism, not shown. And the tip **232** is brought into contact with the base material **230** in a state where the one cutting-edge surface **234***a* diagonally directed to the inner peripheral side (inner side in the radial direction) and gradually cuts the rotating base material **230** while performing the forward/backward movement and the vertical/lateral movement as necessary. The cutting tool **231** is guided so that the closer the tip **232** goes to the outer peripheral side, the deeper it cuts the base material **230**, whereby the inclined portion **221** is formed.

Moreover, the cutting tool **231** is made to go backward with respect to the disc portion **212** while moving to the outer peripheral side. And the tip **232** moves so as to reach the side of the rotor-blade forming portion **217** from the disc portion **212**, whereby the curved portion **222** is formed. The movement of the cutting tool **231** as above is performed in

a narrow width in the radial direction as compared with the formation of the inclined portion **221**. As a result, between the inclined portion **221** and the curved portion **222**, as shown in FIG. **7**, a width in the radial direction (width of an annular part) W**1** in the inclined portion **221** becomes larger than a width in the radial direction (width of the same annular part) W**2** in the curved portion **222**.

As described above, by setting the inclination angle α**1** related to the inclined portion **221** smaller than the tangent angle α**2** related to the curved portion **222**, the groove portion **218** can be formed as smoothly as possible. Moreover, at a part on the outer peripheral side of the disc portion **212**, the base material **230** can be machined by disposing the cutting tool **231** closer to the inner peripheral side, which is the side where there is no rotor-blade forming portion **217**.

And the fabrication of the groove portion **218**, the space (inner space) on the inner peripheral surface side of the base material **230** can be effectively utilized. Moreover, such machining can be realized that the tip **232** does not interfere with the rotor-blade forming portion **217**. As a result, when the groove portion **218** is fabricated, such a work of changing a direction of the cutting tool **231** is not necessary any more, and the groove portion **218** can be fabricated easily with a smaller number of processes. Moreover, the groove portion **218** can be fabricated with the general cutting tool **231** without preparing a dedicated tool.

Subsequently, a stress control function exerted between the rotating body **103** and the rotor shaft **113** will be explained. FIG. **11** illustrates a part surrounded by a one-dot chain line circle D in FIG. **1** in an enlarged manner. Here, in FIG. **11**, the fitting shaft portion **241** of the rotor shaft **113** is not cut vertically but a part on the lower side in the figure of the fitting shaft portion **241** is cut vertically.

In the example in FIG. **11**, at a root part in the fitting shaft portion **241** of the rotor shaft **113**, a stress-control recess portion **251** (counterbore portion) is formed. This stress-control recess portion **251** is formed with a certain depth (approximately 0.1 to 0.5 mm, for example) by counterboring an opening portion of a female screw portion **252** into which the bolt **214** is screwed in the periphery of the root part in the fitting shaft portion **241**. Moreover, the stress-control recess portion **251** is formed so as to face a peripheral edge part of the opening portion of the bolt through hole **213** in the rotating body **103**.

This stress-control recess portion **251** is configured to receive a protruding portion (not shown), if it is present on a surface opposed to the rotating body **103** and the rotor shaft **113**, in a space in the inner side so that the opposed surface does not contact (is not contacted) or press the protruding portion. And the stress-control recess portion **251** prevents the stress generated in the disc portion **212** from increasing by application of a stress generated in the protruding portion or a contact surface with the protruding portion.

As a result, collapse of a relationship between the stress generated in the fitting hole portion **215** or the through hole portion **216** and the stress generated in the groove portion **218** by the pressing of the protruding portion can be prevented. And the stress distribution function of the groove portion **218** is exerted as designed without being affected by the pressing of the protruding portion. Moreover, by providing the stress-control recess portion **251**, the stress distribution function by the groove portion **218** can be made to function more reliably.

Here, as the protruding portion, a substance generated after applying surface processing (electroless nickel plating or the like) (so-called plating drip) to the inner peripheral surfaces of the fitting hole **211** and the bolt through hole **213** for improving resistance against an erosive gas, unexpected protruding part generated on the fitting hole portion **215** and the through hole portion **216** and the like can be exemplified. And even if the protruding portion as above is generated at a part close to the fitting hole **211** or the bolt through hole **213**, by preventing or suppressing stress generation by the stress-control recess portion **251**, the function of the groove portion **218** can be exerted to the maximum.

Note that, in FIG. **11**, the example in which the stress-control recess portion **251** is provided on the rotor shaft **113** is illustrated, but this is not limiting, and as shown in FIG. **12**, the stress-control recess portion **254** (counterbore) may be provided on the side of the rotating body **103**, for example. In the example in FIG. **12**, by counterboring the opening portion of the bolt through hole **213**, it is formed with a certain depth (approximately 0.1 to 0.2 mm, for example). If the stress-control recess portion **254** is formed on the side of the rotating body **103** as above, the working effects of the invention similar to that in the example in FIG. **11** can be exerted.

As components forming the stress-control recess portions **251**, **254**, a component on the side where the protruding part is not generated (or difficult to be generated) can be considered. For example, in the rotating body **103** and the rotor shaft **113**, since the rotor shaft **113** is rarely plated, the stress-control recess portion **251** may be formed on the rotor shaft **113**.

According to the turbo molecular pump **100** of this embodiment as described above, the stress on the through hole portion **216** and the fitting hole portion **215** can be distributed by the groove portion **218** provided on the rotating body **103** at rotation of the rotating body **103**. Thus, the stress that can be generated in the through hole portion **216** and the fitting hole portion **215** is increased and as a result, reliability and performances of the rotating body **103** and the turbo molecular pump **100** can be improved.

The averaging of the stress by the groove portion **218** does not change energy relating to the stress generated in the through hole portion **216**, the fitting hole portion **215**, and the rotating body **103** as a whole. However, if the groove portion **218** is not provided, stress at a part where stress exceeding the average could be generated can be lowered.

Moreover, the groove portion **218** has the inclined portion **221** with the inclination angle $\alpha 1$ and the curved portion **222** with the tangent angle $\alpha 2$, and the inclination angle $\alpha 1$ and the tangent angle $\alpha 2$ have the relationship of $\alpha 1 < \alpha 2$. And it can be also described that the inclined portion **221** is gentler than the curved portion **222**, and the curved portion **222** is steeper than the inclined portion **221**. Thus, the stress can be distributed appropriately for the through hole portion **216** and the fitting hole portion **215** by the gentle inclined portion **221** and the steep curved portion **222**.

Note that the both inclination angle $\alpha 1$ and the tangent angle $\alpha 2$ can similarly have gentle angles. In that case, the similar stress distribution can be generated in the both inclined portion **221** and the curved portion **222**. On the other hand, by causing the curved portion **222** to have a structure steeper than that of the inclined portion **221** as described above, the radial width (width of the annular part) W2 in the curved portion **222** can be made smaller.

Moreover, in the example in FIG. **7**, since the inclined portion **221** is formed on the inner peripheral side away from the rotor-blade forming portion **217**, and the curved portion **222** is formed on the outer peripheral side close to the rotor-blade forming portion **217**, the internal space of the base material **230** can be effectively utilized at machining of the inclined portion **221**, and the groove portion **218** can be machined while preventing interference of the base end side of the tip **232** in the cutting tool **231** with the rotor-blade forming portion **217**. And the machining of the groove portion **218** can be performed with a lower cost.

Here, in the example in FIG. **7**, the sectional shape of the groove portion **218** is constituted by combining the inclined portion **221** and the curved portion **222**, but this is not limiting, and a groove portion **228** may be formed by combining a first inclined portion **226** and a second inclined portion **227** as shown in FIG. **8**, for example. In the example in FIG. **8**, the first inclined portion **226** is formed similarly to the inclined portion **221** in the example in FIG. **7**, but the second inclined portion **227** is constituted not by an arc-shaped surface but a substantially flat inclined surface with an inclination angle $\alpha 3$. Moreover, in the example in FIG. **8**, the first inclined portion **226** and the second inclined portion **227** continue to each other through a connecting curved-surface portion **229** having an arc-shaped sectional shape.

In the example in FIG. **8**, too, it can be described that the first inclined portion **226** on the inner peripheral side is gentler than the second inclined portion **227** on the outer peripheral side and that the second inclined portion **227** is steeper than the first inclined portion **226**. And the groove portion **228** can be machined while effectively utilizing the internal space of the base material **230**.

The embodiment of the present invention has been explained as above, but the present invention is not limited to the aforementioned embodiment but is capable of various variations. For example, the disposition of the groove portion **218** is not limited to the boundary part **224** with respect to the rotor-blade forming portion **217** in the disc portion **212** as shown in FIG. **5**, but it may be disposed on any of parts closer to the inner peripheral side than the boundary part **224** (part on the outer peripheral side of the fitting hole portion **215** or the through hole portion **216** and on the inner peripheral side of the boundary part **224**). Moreover, a plurality of the groove portions **218** may be provided in a space between the boundary part **224** and the fitting hole **211**.

Moreover, the groove portion **218** can be provided on at least either one of the inner peripheral surface and the outer peripheral surface of the rotating body **103**. And the disposition of the groove portion **218** can be configured such that it is opened in the outer peripheral surface of the rotating body **103**. As the outer peripheral surface of the rotating body **103**, an outer peripheral surface **225** of the disc portion **212** can be exemplified. Moreover, as the inner peripheral surface and the outer peripheral surface of the rotating body, the inner peripheral surface and the outer peripheral surface of the cylinder portion **102***d* (FIG. **1**) closer to the exhaust side than the rotor blade **102** in the rotating body **103** can be cited. Furthermore, regarding distinction between the inner peripheral surface and the outer peripheral surface of the rotating body **103**, a surface facing the internal space of the rotating body **103** can be distinguished as the inner peripheral surface from the other as the outer peripheral surface, for example.

Here, the stress distribution function by the groove portion **218** is considered to be exerted more easily by disposing the groove portion **218** closer to the outer peripheral side of the connected part (the fitting hole portion **215** and the through hole portion **216**) between the rotating body **103** and the rotor shaft **113** and a part closer to the connected part. And as the part as above, in the example of FIG. **5**, a part closer to the outer peripheral side than the through hole portion **216** in the inner peripheral surface **219** or the outer peripheral surface **225** of the disc portion **212** can be cited.

Moreover, the groove portion **218** can be provided on both of the inner peripheral surface **219** and the outer peripheral surface **225** of the disc portion **212**. In the analysis by the inventor and the like, the stress generated in the fitting hole portion **215** and the through hole portion **216** was smaller in the case where the groove portion **218** was provided on the inner peripheral surface **219** than the case of being provided on the outer peripheral surface **225**. Moreover, when the groove portion **218** was provided on the both inner peripheral surface **219** and the outer peripheral surface **225** of the disc portion **212** as described above, the stress was averaged, and the effect of the stress distribution was further improved.

Moreover, according to analysis by the inventor and the like, when the thickness H of the disc portion **212** (FIG. **5**) is made smaller, the effect of the stress distribution by provision of the groove portion **218** is prominent. Furthermore, in the examples in FIG. **1** and FIG. **5**, the groove portion **223***a* facing the plate surface of the counterbore **220** is formed on the outer peripheral part of the bottom part in the recessed portion **223**, and this groove portion **223***a* can be also considered to have the stress distribution function.

Moreover, in FIG. **1**, FIG. **5** and the like, with regard to the connection relationship between the rotating body **103** and the rotor shaft **113**, such a type that the rotor shaft **113** is inserted so as to penetrate the fitting hole **211** of the rotating body **103** is exemplified, but the present invention is not limited to that, and as shown in FIGS. **13**(*b*) to **13**(*d*), for example, it can be also applied to various types of rotating bodies **103***b* to **103***d* as shown in partially vertical sections.

For example, FIG. **13**(*a*) illustrates a part of the rotating body **103** according to the embodiment shown in FIG. **5** (disc portion **212**) in an enlarged manner as a partially and vertical section, but this is not limiting, and as shown in FIG. **13**(*b*), the present invention can be applied also to the type of the rotating body **103***b* which does not include the fitting hole (reference numeral **211** in FIG. **5**) and to which the rotor shaft **113***b* is connected. The rotor shaft **113***b* shown in FIG. **13**(*b*) is the one not including the fitting shaft portion **241** or the protruding end portion **242** in the example shown in FIG. **13**(*a*). As described above, for the structure in which the rotor shaft **113***b* abuts against the rotating body **103***b* so as to fasten the both, the groove portion **218** is disposed on the fastened surface of the rotating body **103***b* (an inner peripheral surface **219***b* against which the rotor shaft **113***b* abuts) or an extended surface thereof.

Moreover, the one shown in FIG. **13**(*c*) is a type of the rotating body **103***c* not having the bolt hole around the fitting hole **211***c* in the disc portion **212***c*. This type of the rotating body **103***c* is connected to the rotor shaft **113***c* such that a nut **256** is attached to a protruding end portion **242***c* in the rotor shaft **113***c*, and the counterbore **220** is pressed onto the rotating body **103***c* by fastening the nut **256**.

Furthermore, the one shown in FIG. **13**(*d*) is the type of the rotating body **103***d* in which the fitting hole **211***d* does not penetrate the disc portion **212***d* and is closed at a part in the middle of the thickness direction of the disc portion **212***d*. When this type of rotating body **103***d* is employed, the rotor shaft **113***d* is fixed to the rotating body **103***d* by the bolt **214** in a state where the rotor shaft **113***d* does not penetrate the disc portion **212***d*. Note that the rotating body **103***d* and the rotor shaft **113***d* shown in FIG. **13**(*d*) can be considered as the one in which a protruding portion **257** formed on an end part in the axial direction of the rotor shaft **113***d* is inserted into a recess portion (reference numeral omitted) of the rotating body **103***d* so as to engage the rotor shaft **113***d* with the rotating body **103***d*.

Moreover, the one shown in FIG. **14**(*a*) is a rotating body **103***e*, which is similar to the rotating body **103***b* of the type shown in FIG. **13**(*b*) but is different in a point that it has an engagement structure by a protrusion **258** between it and a counterbore **220***e*. In the example in FIG. **14**(*a*), the protrusion **258** is formed on the counterbore **220***e*, and this protrusion **258** enters a recess part (reference numeral omitted) of the rotating body **103***e*. As described above, the structure which causes the rotor shaft **113***b* to abut against the rotating body **103***b* so as to fasten the both, too, similarly to the example in FIG. **13**(*b*), the groove portion **218** is disposed on the fastened surface of the rotating body **103***e* (an inner peripheral surface **219***e* against which the rotor shaft **113***e* abuts) or on the extended surface thereof.

Furthermore, the one shown in FIG. **14**(*b*) is a type of a rotating body **103***f* having a protruding portion **259** and engaged with a rotor shaft **113***f* by inserting this protruding portion **259** into a recess part (reference numeral omitted) of the rotor shaft **113***f*.

Furthermore, the one shown in FIG. **14**(*c*) is a type of a rotating body **103***g* having a protrusion **260** and the protruding portion **259**, the protrusion **260** is caused to enter a recess part of the counterbore **220***g*, and the protruding portion **259** is inserted into a recess part (reference numeral omitted) of the rotor shaft **113***g* similarly to the example in FIG. **14**(*b*).

In the turbo molecular pump including these various types of the rotating bodies **103***b* to **103***g* and the rotor shafts **113***b* to **113***g* and the like as described above, too, by providing the groove portions **218** and **223***a* at appropriate positions, the stress distribution function can be exerted similarly to the turbo molecular pump **100** shown in FIG. **1**, FIG. **5** and the like.

Note that the present invention is not limited to the aforementioned embodiment but is capable of many variations by ordinary creative capabilities of a person ordinarily skilled in the art as long as they are within the technical scope of the present invention.

Although elements have been shown or described as separate embodiments above, portions of each embodiment may be combined with all or part of other embodiments described above.

Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are described as example forms of implementing the claims.

The invention claimed is:

1. A vacuum pump having a rotating blade on an outer periphery of a cylinder part and including a rotating body fastened to a rotating shaft by a fastener and rotatable together with the rotating shaft, wherein

in the rotating body, at least one of a fitting hole portion fitted with the rotating shaft and a through hole portion, through which the fastener penetrates, is a stress-reduction target portion, and a groove portion that reduces stress generated in the stress-reduction target portion during rotation of the rotating body is provided in the rotating body,

the rotating body has a recessed portion, and

the groove portion consists of a first groove portion provided adjacent a planar inner peripheral surface of the rotating body and a second groove portion formed on a bottom part in the recessed portion, wherein the planar inner peripheral surface extends within a geometric plane that includes a portion that extends across the first groove portion and wherein the first groove portion comprises an inclined surface extending from the planar inner peripheral surface at an acute angle relative to the portion of the geometric plane that extends across the first groove portion,

wherein the first groove portion comprises an inner peripheral side and an outer peripheral side, wherein the inner peripheral side comprises the inclined surface and the acute angle of the inclined surface is smaller than an angle at which a surface of the outer peripheral side intersects with the portion of the geometric plane that extends across the first groove portion.

2. The vacuum pump according to claim **1**, wherein

the groove portion is provided closer to an outer peripheral side than the fitting hole portion or the through hole portion.

3. The vacuum pump according to claim **1**, wherein

the first groove portion is disposed on a fastened surface of the rotating body or on an extended surface thereof.

4. The vacuum pump according to claim **1**, wherein

the rotating body is applied with surface treatment, and has a counterbore portion, which avoids contact with the rotating shaft, on a peripheral edge part of at least one of the fitting hole portion and the through hole portion.

5. The vacuum pump of claim **1** wherein the inclined surface extends from the planar inner peripheral surface to the outer peripheral side of the first groove portion.

6. The vacuum pump of claim **5** wherein the rotating shaft rotates about an axis of rotation and wherein the inner peripheral side of the first groove portion has a width along the geometric plane in a radial direction relative to the axis of rotation and the outer peripheral side of the first groove portion has width along the geometric plane in the radial direction, the width of the inner peripheral side being greater than the width of the outer peripheral side.

7. The vacuum pump of claim **6** wherein the outer peripheral side comprises a curved surface extending from the inclined surface of the inner peripheral side to the geometric plane.

8. The vacuum pump of claim **6** wherein the outer peripheral side comprises an inclined surface and a curved surface, the inclined surface of the outer peripheral side extending from the geometric plane to the curved surface and the curved surface extending from the inclined surface of the outer peripheral side to the inclined surface of the inner peripheral side.

9. A rotating body for a vacuum pump having a rotor blade on an outer periphery of a cylinder part and fastened to a rotating shaft by a fastener, wherein

at least one of a fitting hole portion fitted with the rotating shaft and a fastened part, to which the fastener is fastened, is a stress-reduction target portion, the rotating body including a groove portion which reduces stress generated in the stress-reduction target portion during rotation,

the rotating body has a recessed portion, and

the groove portion consists of a first groove portion provided adjacent a planar inner peripheral surface of the rotating body and a second groove portion formed on a bottom part in the recessed portion, wherein the planar inner peripheral surface extends within a geometric plane that includes a portion that extends across the first groove portion and wherein the first groove portion comprises an inclined surface extending from the planar inner peripheral surface at an acute angle relative to the portion of the geometric plane that extends across the first groove portion,

wherein the first groove portion comprises an inner peripheral side and an outer peripheral side, wherein the inner peripheral side comprises the inclined surface and the acute angle of the inclined surface is smaller than an angle at which a surface of the outer peripheral side intersects with the portion of the geometric plane that extends across the first groove portion.

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